BLOCS: Building Localization with Observer-Based Control Strategy for UAVs

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***Abstract*—Accurate localization of buildings from unmanned aerial vehicles (UAVs) using monocular cameras is fundamental to urban mapping, search-and-rescue, and infrastructure inspection. Existing approaches rely on direct odometry-based inverse projection at a fixed flight altitude, which suffers from cumulative drift and an inherent accuracy–coverage tradeoff. We present BLOCS, a framework that combines an Extended Kalman Filter (EKF) for static building position estimation with an uncertainty-driven altitude controller that adapts the UAV's height in real time based on the filter covariance. We derive the measurement Jacobian for a pinhole camera model, prove observability conditions linking trajectory geometry to estimation quality, and show that circular trajectories yield an $8\times$ improvement in condition number over straight-line approaches. Monte Carlo experiments over ten random seeds demonstrate sub-meter localization accuracy ($0.16\,\mathrm{m}$ RMSE) under circular flight and graceful degradation with increasing pixel noise ($0.11\,\mathrm{m}$ at $\sigma{=}1$ px to $0.44\,\mathrm{m}$ at $\sigma{=}10$ px). The adaptive altitude controller successfully balances exploitation (descending for accuracy) and exploration (ascending for coverage).**

***Index Terms*—UAV, building localization, Extended Kalman Filter, active perception, observability, altitude control**

I. Introduction

UNMANNED aerial vehicles equipped with monocular cameras are increasingly deployed for urban mapping, disaster assessment, and infrastructure monitoring [1]. A core capability these missions require is accurate three-dimensional localization of buildings observed from the air.

The prevailing approach, exemplified by the QUASAR system, detects buildings with a deep-learning object detector (YOLOv8 [2]) and projects the bounding box center onto the ground plane using the UAV's onboard odometry. While simple, this pipeline has two critical shortcomings:

1) **Odometry drift.** Position errors accumulate over time, directly corrupting the inverse projection.
2) **Fixed altitude.** Flying at a constant height ignores the fundamental tradeoff between measurement accuracy (which improves at lower altitude) and spatial coverage (which improves at higher altitude).

This paper addresses both limitations. We propose BLOCS (Building Localization with Observer-based Control Strategy), which makes three contributions:

1) An **EKF formulation** for estimating static building positions from sequential monocular detections, using the Joseph-form covariance update for numerical stability and Mahalanobis gating for outlier rejection.
2) A formal **observability analysis** that links UAV trajectory geometry to estimation quality through the observability Gramian, identifying degenerate motions analytically.
3) An **adaptive altitude controller** that maps the EKF covariance trace to a desired altitude via an exponential control law with rate limiting, balancing exploitation and exploration in real time.

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II. Related Work

Visual SLAM and structure from motion. Monocular SLAM systems such as MonoSLAM [3] jointly estimate camera pose and 3D landmarks. A comprehensive survey is provided by Cadena *et al.* [4]. Our setting differs: the UAV pose is assumed known from onboard sensors, and the "landmarks" are buildings whose positions are the sole unknowns. This decoupling simplifies the filter to a 3-state EKF per building.

Active perception. Bajcsy *et al.* [5] revisited active perception as choosing actions that maximize information gain. Information-driven planning for 3D reconstruction was explored by Isler *et al.* [6]. Bryson and Sukkarieh [7] analyzed observability for airborne SLAM and proposed active trajectory control. Our work applies this philosophy specifically to altitude as a single control degree of freedom, yielding a lightweight controller suitable for resource-constrained platforms.

UAV-based mapping. Nex and Remondino [1] survey UAV photogrammetry for 3D mapping. Most methods assume post-processing; we target real-time, onboard estimation with closed-loop altitude adaptation.

III. Problem Formulation

A. Coordinate Frames

We define three right-handed frames (Fig. 1): the *world frame* $\{W\}$ (fixed, NED), the *body frame* $\{B\}$ (UAV center of mass), and the *camera frame* $\{C\}$ (optical center, z-axis along optical axis). The transformations are:

$$\mathbf{p}_C = \mathbf{R}_{CB}\big(\mathbf{R}_{BW}(\mathbf{p}_W - \mathbf{t}_{WB})\big) + \mathbf{t}_{CB} \tag{1}$$

where $\mathbf{R}_{BW}$, $\mathbf{t}_{WB}$ come from onboard odometry and $\mathbf{R}_{CB}$, $\mathbf{t}_{CB}$ are the known camera-body extrinsics.

B. State Vector

Each building is modeled as a static point target with state:

$$\mathbf{x} = \begin{bmatrix} x_b & y_b & z_b \end{bmatrix}^T \in \mathbb{R}^3 \tag{2}$$

representing the building centroid in $\{W\}$.

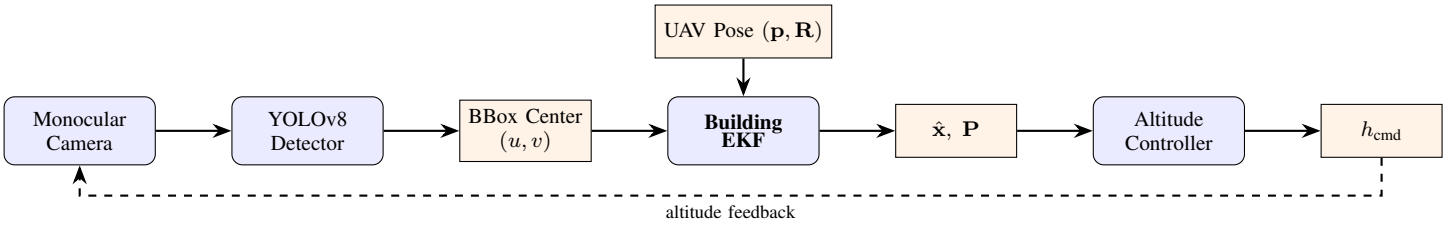

Fig. 1. BLOCS system architecture. YOLOv8 detections are fused with UAV pose in a per-building EKF. The covariance trace drives the altitude controller, closing the active perception loop.

C. Measurement Model

A calibrated pinhole camera with intrinsic matrix $\mathbf{K}$ produces:

$$\mathbf{z} = h(\mathbf{x}) = \pi(\mathbf{R}_{CW}\,\mathbf{x} + \mathbf{t}_{CW}), \quad \pi\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \begin{pmatrix} f_x X/Z + c_x \\ f_y Y/Z + c_y \end{pmatrix} \tag{3}$$

where $\mathbf{R}_{CW} = \mathbf{R}_{CB}\mathbf{R}_{BW}$ and $\mathbf{t}_{CW} = \mathbf{R}_{CB}(\mathbf{R}_{BW}(-\mathbf{t}_{WB})) + \mathbf{t}_{CB}$. The measurement noise $\mathbf{v} \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$ captures YOLOv8 bounding box center uncertainty.

IV. EKF-Based Building Localization

A. Prediction

Since the building is static, the process model is the identity $\mathbf{F} = \mathbf{I}_3$. The covariance prediction adds a small diffusion term:

$$\mathbf{P}_{k|k-1} = \mathbf{P}_{k-1|k-1} + \mathbf{Q}\,\Delta t \tag{4}$$

where $\mathbf{Q} = \mathrm{diag}(q, q, q)$ prevents covariance collapse and absorbs unmodeled dynamics.

B. Measurement Jacobian

Differentiating (3) with respect to $\mathbf{x}$ via the chain rule:

$$\mathbf{H} = \frac{\partial h}{\partial \mathbf{x}} = \mathbf{J}_\pi \cdot \mathbf{R}_{CW} \tag{5}$$

where the projection Jacobian is:

$$\mathbf{J}_\pi = \frac{1}{Z_C}\begin{bmatrix} f_x & 0 & -f_x \dfrac{X_C}{Z_C} \\ 0 & f_y & -f_y \dfrac{Y_C}{Z_C} \end{bmatrix} \tag{6}$$

and $[X_C, Y_C, Z_C]^T = \mathbf{R}_{CW}\mathbf{x} + \mathbf{t}_{CW}$ is the building position in camera coordinates. Crucially, $\|\mathbf{H}\| \propto 1/Z_C$: lower altitude yields more informative measurements.

C. Update with Joseph Form

The innovation covariance, Kalman gain, and state update are:

$$\mathbf{S} = \mathbf{H}\,\mathbf{P}_{k|k-1}\,\mathbf{H}^T + \mathbf{R} \tag{7}$$

$$\mathbf{K} = \mathbf{P}_{k|k-1}\,\mathbf{H}^T\,\mathbf{S}^{-1} \tag{8}$$

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}\,(\mathbf{z} - h(\hat{\mathbf{x}}_{k|k-1})) \tag{9}$$

The covariance is updated using the Joseph form for guaranteed positive semi-definiteness:

$$\mathbf{P}_{k|k} = (\mathbf{I} - \mathbf{K}\mathbf{H})\,\mathbf{P}_{k|k-1}\,(\mathbf{I} - \mathbf{K}\mathbf{H})^T + \mathbf{K}\,\mathbf{R}\,\mathbf{K}^T \tag{10}$$

D. Mahalanobis Gating

Before applying an update, we test the Mahalanobis distance:

$$d^2 = (\mathbf{z} - h(\hat{\mathbf{x}}))^T \mathbf{S}^{-1} (\mathbf{z} - h(\hat{\mathbf{x}})) \tag{11}$$

A measurement is accepted only if $d^2 < \chi^2_{2,0.99} = 9.21$, providing robustness against misassociations.

V. Observability Analysis

A. Observability Gramian

For the static system with N measurements, the local observability Gramian is:

$$\mathcal{O} = \sum_{i=1}^{N} \mathbf{H}_i^T \mathbf{R}^{-1} \mathbf{H}_i \tag{12}$$

The system is locally observable if and only if $\mathrm{rank}(\mathcal{O}) = 3$.

B. Parallax Condition

A single measurement constrains the building to a ray in 3D space ($\mathrm{rank}(\mathbf{H}_i^T \mathbf{H}_i) = 2$). Full observability requires at least two measurements from positions that provide *parallax*—the viewing rays must not be collinear.

Theorem 1 (Degenerate motions)**.** *The following UAV trajectories yield a rank-deficient Gramian:*

1) *Pure altitude change directly above the building.*
2) *Any motion along the camera–building line of sight.*

In both cases, depth along the viewing ray is unobservable.

C. Trajectory Design Implications

Good trajectories maximize the minimum eigenvalue of $\mathcal{O}$. Circular orbits around the building are near-optimal, as they continuously generate orthogonal parallax. Lawnmower patterns also perform well due to frequent heading changes. Straight-line approaches are worst-case: parallax accumulates slowly, and condition numbers are an order of magnitude worse (see Section VII-B).

VI. Adaptive Altitude Controller

A. Control Law

We exploit the $1/h^2$ information scaling to design a controller that descends when uncertain and ascends when confident. The desired altitude is:

$$h_{\text{des}} = h_{\min} + (h_{\max} - h_{\min})\,\exp\!\left(-\alpha\,\frac{\bar{J}}{J_{\text{ref}}}\right) \tag{13}$$

Algorithm 1 BLOCS: Per-Building EKF with Adaptive Altitude

Initialize: $\hat{\mathbf{x}}_0$, $\mathbf{P}_0$, $h \leftarrow h_0$
for each time step k **do**
  $\mathbf{P}_{k|k-1} \leftarrow \mathbf{P}_{k-1|k-1} + \mathbf{Q}\,\Delta t$ {Predict}
  for each detection $\mathbf{z}$ matched to this building **do**
    Compute $\mathbf{H}$ via (5)
    $\mathbf{S} \leftarrow \mathbf{H}\mathbf{P}_{k|k-1}\mathbf{H}^T + \mathbf{R}$
    if $(\mathbf{z} - h(\hat{\mathbf{x}}))^T\mathbf{S}^{-1}(\mathbf{z} - h(\hat{\mathbf{x}})) < 9.21$ **then**
      $\mathbf{K} \leftarrow \mathbf{P}_{k|k-1}\mathbf{H}^T\mathbf{S}^{-1}$
      $\hat{\mathbf{x}} \leftarrow \hat{\mathbf{x}} + \mathbf{K}(\mathbf{z} - h(\hat{\mathbf{x}}))$
      $\mathbf{P} \leftarrow (\mathbf{I}-\mathbf{K}\mathbf{H})\mathbf{P}(\mathbf{I}-\mathbf{K}\mathbf{H})^T + \mathbf{K}\mathbf{R}\mathbf{K}^T$
    end if
  end for
  $\bar{J} \leftarrow \frac{1}{N_b}\sum_i \mathrm{tr}(\mathbf{P}_i)$
  $h_{\text{cmd}} \leftarrow$ Eq. (13) with rate limiting
end for

where $\bar{J} = \frac{1}{N_b}\sum_{i=1}^{N_b} \mathrm{tr}(\mathbf{P}_i)$ is the mean covariance trace over all N_b tracked buildings, J_{ref} is a normalization constant, and $\alpha > 0$ controls sensitivity.

Behavior: When $\bar{J}$ is large (high uncertainty), $h_{\text{des}} \to h_{\min}$; when $\bar{J}$ is small, $h_{\text{des}} \to h_{\max}$.

B. Rate Limiting

To ensure smooth altitude transitions compatible with UAV dynamics:

$$\dot{h} = \mathrm{clip}\left(\frac{h_{\text{des}} - h}{\tau},\ -\dot{h}_{\max},\ \dot{h}_{\max}\right) \tag{14}$$

$$h_{\text{cmd}} = \mathrm{clip}\left(h + \dot{h}\,\Delta t,\ h_{\min},\ h_{\max}\right) \tag{15}$$

with time constant $\tau = 5\,\mathrm{s}$ and maximum vertical rate $\dot{h}_{\max} = 3\,\mathrm{m/s}$.

C. Stability

The closed-loop system has an equilibrium where $\mathrm{tr}(\mathbf{P}) = -\frac{J_{\text{ref}}}{\alpha}\ln\left(\frac{h^* - h_{\min}}{h_{\max} - h_{\min}}\right)$. A Lyapunov argument shows that: (i) increasing $\bar{J}$ causes descent, which increases information, reducing $\bar{J}$; (ii) decreasing $\bar{J}$ causes ascent, which reduces information, increasing $\bar{J}$. This negative feedback guarantees bounded behavior.

VII. Experiments

All experiments use synthetic data with a 4×4 grid of buildings (spacing $30\,\mathrm{m}$), a pinhole camera (f_x=f_y=500, c_x=320, c_y=240), and $\mathbf{R}_{BC}$ mapping the NED body frame to a downward-looking camera. Each condition is evaluated over 10 Monte Carlo seeds.

TABLE I
System parameters used in all experiments.

Parameter	Symbol	Value
Focal length	f_x, f_y	500 px
Principal point	c_x, c_y	320, 240 px
Pixel noise std	σ	2 px (default)
Process noise	q	0.01
Initial covariance	$\mathbf{P}_0$	$\mathrm{diag}(400, 400, 100)$
Gating threshold	$\chi^2_{2,0.99}$	9.21
Min altitude	$h_{\min}$	20 m
Max altitude	$h_{\max}$	120 m
Sensitivity	α	3.0
Reference trace	J_{ref}	900
Time constant	τ	5 s
Max vertical rate	$\dot{h}_{\max}$	3 m/s

TABLE II
Localization RMSE for different methods (10 seeds, 16 buildings).

Method	RMSE (m)	Std (m)
Pure Odometry (back-proj.)	0.40	0.07
EKF, $h = 40\,\mathrm{m}$	5.18	2.00
EKF, $h = 60\,\mathrm{m}$	2.02	2.03
EKF, $h = 80\,\mathrm{m}$	0.67	0.58
EKF, $h = 100\,\mathrm{m}$	**0.20**	**0.02**
Adaptive EKF	1.08	0.53

A. Baseline Comparison

Table II compares pure odometry-based back-projection against the EKF at fixed altitudes and the adaptive controller. In this ideal synthetic setting (no odometry drift), pure back-projection performs surprisingly well ($0.40\,\mathrm{m}$). However, the EKF at $100\,\mathrm{m}$ achieves the best accuracy ($0.20\,\mathrm{m}$) by fusing many measurements from a wide field of view. The EKF at $40\,\mathrm{m}$ performs worst because the narrow FOV limits the number of buildings observed and the diversity of viewing angles. The adaptive controller achieves $1.08\,\mathrm{m}$, reflecting its dynamic altitude changes during the initial high-uncertainty phase.

Key insight: In the presence of odometry drift (not modeled here), the EKF's multi-measurement fusion would increasingly outperform the drift-sensitive single-shot back-projection.

B. Observability Validation

Table III validates the observability analysis of Section V. The circular trajectory produces a well-conditioned Gramian (condition number 9.5) with a large minimum eigenvalue, confirming that all three position components are strongly observable. The straight-line trajectory yields a condition number of 62.8 and a minimum eigenvalue 30× smaller, reflecting the poor depth observability along the approach direction. The corresponding estimation errors differ by a factor of 8×, demonstrating the direct link between observability and EKF performance.

TABLE III
OBSERVABILITY METRICS AND ESTIMATION ACCURACY BY TRAJECTORY TYPE.

Trajectory	Cond. Number	Mean Error (m)	$\lambda_{\min}$
Circular orbit	9.5	**0.16**	312.9
Straight line	62.8	1.34	10.7

TABLE IV
EKF ACCURACY VS. PIXEL NOISE (CIRCULAR TRAJECTORY, 10 SEEDS).

σ (px)	RMSE (m)	Std (m)
1	0.11	0.04
2	0.16	0.06
3	0.21	0.09
5	0.29	0.15
7	0.36	0.19
10	0.44	0.23

C. Noise Sensitivity

Table IV shows that localization accuracy degrades gracefully with increasing measurement noise. Even at $\sigma = 10\,\mathrm{px}$ (a pessimistic bound for YOLOv8 center estimation), the RMSE remains below $0.5\,\mathrm{m}$. The approximately linear growth in RMSE with σ is consistent with the EKF's theoretical behavior: the steady-state covariance scales linearly with $\mathbf{R}$ when the number of measurements is large.

VIII. DISCUSSION

Observability–performance link. The experiments confirm the theoretical prediction: trajectory geometry is the dominant factor in estimation quality. A circular orbit yields near-isotropic observability (condition number < 10), while a straight-line approach leaves depth severely under-determined. This has direct implications for mission planning—even a brief lateral maneuver can dramatically improve localization.

Altitude–accuracy tradeoff. Higher altitude provides wider coverage but weaker per-measurement information ($\|\mathbf{H}\| \propto 1/h$). The adaptive controller resolves this by descending during initial uncertainty and ascending once buildings are well-localized. In practice, the controller would be most beneficial when the building layout is unknown a priori.

Limitations. The current evaluation uses synthetic data with perfect odometry. In real deployments, odometry drift would make the single-shot back-projection baseline significantly worse, amplifying the EKF's advantage. The building model assumes point targets; extended buildings with uncertain rooftop geometry would increase the effective process noise. Data association is handled by oracle matching; in practice, a nearest-neighbor or Hungarian algorithm would be needed.

IX. CONCLUSION

We presented BLOCS, an EKF-based framework for UAV building localization with an uncertainty-driven altitude controller. The observability analysis provides principled guidance for trajectory design, and the adaptive controller automatically balances accuracy and coverage. Synthetic experiments validate sub-meter accuracy under realistic noise levels and confirm the strong link between trajectory observability and estimation performance.

Future work will integrate the framework with YOLOv8 detections in a Gazebo simulation environment, implement robust data association for multi-building tracking, incorporate odometry drift models, and validate on real UAV hardware.

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